CanConstructNZ

Balance in the sector

HELP

User manual

A practical guide to creating scenarios, forecasting construction demand, running the allocation simulation, and interpreting the results.

VIDEO Smart System Dashboard video guide

Watch this practical walkthrough to learn how to enter scenario assumptions, run forecasts and simulations, and interpret the dashboard results.

Watch the dashboard guide on YouTube ↗

Quick start

The system answers two linked questions: *how much construction work might exist?* and *can the sector deliver it?* Always forecast first, then simulate allocation.

1. **Build a scenario**
   Set the economic assumptions, policy changes, shock, and horizon in Pipeline.
2. **Run the forecast**
   Generate the regional pipeline by year and construction type.
3. **Run the simulation**
   Allocate forecast projects to the available main contractors and subcontractors.
4. **Read the results**
   Use Overview and Bottlenecks to find unmet workload and capacity pressure.

01 Enter scenario assumptions

Open **Pipeline**. The left rail contains all scenario controls.

ITEM	WHAT IT MEANS	HOW TO INTERPRET IT
GDP growth (annual)	The assumed annual change in economic output.	Higher GDP growth generally raises forecast construction activity; negative or weak growth generally suppresses it. Treat it as a scenario assumption, not a prediction.
Population growth (annual)	The assumed annual change in population.	Higher growth generally increases demand, especially for residential and population-serving buildings. Effects vary because the model uses fitted historical relationships.
Inflation (development only)	Annual price growth applied where the control is enabled.	Higher inflation can raise nominal NZD values without implying the same increase in real building volume or project count.
Fiscal policy	Limited or Expanded government spending setting.	Expanded represents stronger public investment conditions; compare it with Limited to understand sensitivity rather than reading either as certainty.
Horizon year	The final year of the forecast and the year used for the linked simulation.	A longer horizon compounds assumptions for longer and therefore increases uncertainty.

Policies

Select **+ Add policy**, open the policy card, set its start year, and save it. Up to one of each available policy can be included.

ITEM	WHAT IT MEANS	HOW TO INTERPRET IT
Consent reform	A 10% step-up from the chosen year across every region and building type.	The effect persists from the policy year onward. Compare an otherwise identical scenario without the policy to isolate its modelled effect.
Densification	A temporary residential boost in selected regions: a one-year lag, two-year peak, then a three-year decline.	It affects only selected regions and residential work. At least one region must be selected before the scenario can run.

Shocks and stresses

Select **+ Add shock**, choose the affected region and start year, then save. The current stress test is an earthquake profile.

ITEM	WHAT IT MEANS	HOW TO INTERPRET IT
Earthquake	A fixed nine-year empirical profile from the start year through year + 8.	Expect an initial disruption followed by reconstruction effects in the epicentre region, plus milder spillovers elsewhere. The shaded chart band identifies the affected period.

Good comparison practice
Change one assumption at a time and retain a baseline scenario. Differences between runs then have a clear explanation. Combined GDP, population, policy, and shock changes are valid, but their effects are harder to attribute.

02 Run and read the forecast

Click **Run scenario**. When complete, use **Show scenario** to reopen cached results.

1. Use **Value (NZD M)** to view forecast construction value or **Project count** to view the estimated number of projects.
2. Use the building-type legend to show or hide Residential, Institutional, Commercial, Industrial, Transportation, Utilities, and Site development and earthworks.
3. Read each regional stacked chart to see both the total trajectory and the composition of work.
4. Hover over the chart for year-specific values. Policy and shock bands show when scenario interventions apply.

Value and count tell different stories
A rise in value with a smaller rise in count can mean larger or more expensive projects; it does not necessarily mean the same increase in physical construction. Compare both views.

Past forecast in the header reopens saved scenarios. Choose a forecast. If a completed simulation exists, choose its year and select **Show results**.

03 Inspect the sector

The **Sector** tab describes the supply side: companies and their available annual capacity.

Switch between **Total revenue** (used as the sector-capacity measure) and **Company count**. Filter categories and select regions on the map or league table. The trend and comparison cards help identify where capacity is concentrated and how the synthetic company population changes over time.

Sector is not an allocation result
This page describes the forecast company population and capacity available to the model. To see how much capacity is actually consumed, run the simulation and use Overview or Bottlenecks.

04 Run the allocation simulation

Return to **Pipeline** and click **Run simulation** after the forecast is complete.

The simulation turns the forecast into projects and attempts to allocate each project's annual workload to eligible main contractors and subcontractors. Matching considers construction category, operating region, project scale, and remaining firm capacity. Large projects are considered first.

A run may be queued when workers are busy. Leave the page open and follow the progress panel; once complete, click **Show simulation**. Results open in Overview automatically.

Forecast versus simulation

The forecast estimates demand. The simulation tests deliverability under the modelled firm population and matching rules. A strong pipeline can therefore coexist with a high delivery deficit.

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Read Overview and Bottlenecks

These tabs report the allocation outcome for the selected run and year.

ITEM	WHAT IT MEANS	HOW TO INTERPRET IT
Unassigned projects	Projects with no eligible main contractor.	A high number signals a main-contractor matching or capacity constraint. Use regional/category views and project traces to locate the cause.
Unallocated Project Workload	Annual project workload not awarded to a main contractor or subcontractor.	This is the clearest monetary measure of the delivery gap. Compare it with total annual workload, not total lifetime project value.
With main contractor	Projects successfully matched to a primary contractor.	This does not guarantee all subcontracted workload was placed; also inspect partial and unallocated workload.
Main / subcontractor workload	The shares of annual workload performed in-house by main firms and passed to subcontractors.	The split reflects firm-size-dependent in-house rules. Neither channel is inherently good or bad; the concern is workload left unmatched.
Capacity utilised	Workload booked against firms' start-of-year headline annual capacity.	High utilisation means less resilience for additional work. Low national utilisation can still coexist with local or category-specific shortages.
Capacity remaining	Headline annual capacity left after allocation.	Spare capacity is useful only when its category, geography, and scale fit the unmatched projects.

On **Overview**, switch among **Deficit & unassigned**, **Project annual workload split**, and **Firm capacity split**. Select a map region to filter the report. The largest unassigned projects and firms with remaining capacity appear lower on the page.

On **Bottlenecks**, read the region-by-category heatmap and the utilisation-by-size-band table. Darker or larger problem cells indicate concentrated unassigned work. High utilisation in a size band suggests those firms are saturated; spare capacity in another band may not be suitable for the same projects.

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Interpret results

Use the system for structured comparison and stress testing.

- **Compare scenarios, not just point estimates.** The most useful result is often the change from a consistent baseline.
- **Check geography and category.** Aggregate spare capacity can hide shortages in a particular region, trade, or project scale.

- **Separate nominal value from project count.** Price growth can lift dollar values without an equivalent lift in workload volume.
- **Distinguish unassigned from partial.** Unassigned means no main contractor; partial means some work was placed but a remainder could not be matched.
- **Use the selected year carefully.** Multi-year views may show projects active in other years, while allocation KPIs apply to the simulation year identified on screen.
- **Do not treat a model run as a promise.** Results depend on scenario assumptions, fitted historical relationships, synthetic projects and companies, and allocation rules.

A useful diagnostic pattern

High unallocated workload plus high utilisation in the same region/category indicates a likely capacity shortage. High unallocated workload alongside apparent spare capacity usually points to a mismatch in category, operating area, or project scale.